



COMSATS University Islamabad

Department of Computer Science

Course Description Form (CDF)

Course Information

Course Code: **CSC461**

Credit Hours: **3 (3,0)**

Lab Hours/Week: **0**

Course Title: **Introduction to Data Science**

Lecture Hours/Week: **3**

Pre-Requisites: **MTH262**

Catalogue Description:

This course provides the basic concepts and principles of Data Science. The topics include: Overview of Data Science; Landscape; Tools & Applications; Introduction to R & RStudio; Data Visualization Techniques; Data Wrangling; Statistical Modeling; Predictive Modeling; and Data Science Ethics.

Unit wise Major Topics:

Unit	Topic	No. of teaching hours
1.	Data Science: Overview, Landscape, Tools and Applications.	4.5
2.	Introduction to R & RStudio: Fundamental Structure & Objects, Canonical Data Graphics in R, Statistical Analysis with R, and Applications of Machine Learning in Data Science using R.	10.5
3.	Data Visualization: Composing Data Graphics, Importance of Graphics, Creating Effective Presentation; and Grammar for Graphics.	6
4.	Data Wrangling: Grammar for Data Wrangling; Data Wrangling on Single & Multiple Tables, Tidy Data, and Iteration.	4.5
5.	Statistical Inference: Sampling, Probability Distributions, Statistical Modeling, and Fitting a Model.	6
6.	Predictive Modeling, Classification Models, Clustering, and Evaluating Models.	10.5
7.	Data Science Ethics: Introduction, Truthful falsehoods, Role of Data Science in Society, Some Setting for Professional Ethics, Principles to Guide Ethical Actions, Algorithmic Bias, Data & Disclosure, Reproducibility, and Professional Guidelines for Ethical Conduct.	3
Total Contact Hours		45

Mapping of CLOs and SOs

Sr.#	Unit #	Course Learning Outcomes	Blooms Taxonomy Learning Level	SO
CLO-1	1	Outline fundamental concepts of data science.	<i>Understanding</i>	1
CLO-2	2	Perform data processing and manipulation techniques in R language.	<i>Applying</i>	1,2,4
CLO-3	3-4	Apply visualization and wrangling techniques to different types of datasets.	<i>Applying</i>	1,2,4

CLO-4	5-6	Apply statistical and predictive modeling techniques using R language.	Applying	1,2,4	
CLO -5	7	Infer ethical considerations for data science.	Understanding	7,8	
CLO Assessment Mechanism					
Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4	CLO-5
Quizzes	Quiz 1	Quiz 2	Quiz 3	Quiz 4	-
Assignments	-	Assignment 1	Assignment 2	Assignment 3&4	-
Mid Term Exam	Mid Term Exam	Mid Term Exam	Mid Term Exam	-	-
Final Term Exam	Final Term Exam				
Text and Reference Books					
Textbook:					
1. Modern Data Science with R, Benjamin S. Baumer, Daniel T. Kaplan, Nicholas J. Horton, Chapman and Hall, 2021.					
Reference Book:					
1. Introduction to Data Science Data Analysis and Prediction Algorithms with R, Rafael A. Irizarry, CRC Press, 2021.					